

Total No. of Questions : 8]

SEAT No. :

PE4298

[Total No. of Pages : 2

[6582]-71

S.E. (Artificial Intelligence & Machine Learning)

OBJECT ORIENTED PROGRAMMING

(2019 Pattern) (Semester - III) (218544)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is Constructor? Explain with example. [6]
b) Write a program to calculate an area of circle using parameterized constructor. [6]
c) Explain different types of Constructors in OOP. [6]

OR

- Q2)** a) Explain concept of symbolic constant with example. [6]
b) Differentiate between Constructor and Destructor. [6]
c) Write Pseudo code for Constructor Overloading. [6]

- Q3)** a) Write java program to implement Multiple inheritance. [5]
b) Explain Method overriding with example. [6]
c) Define Inheritance. What are the types of Inheritance? [6]

OR

- Q4)** a) Write a program for multiplication of two numbers using Method overloading. [5]
b) What is abstract class and explain with example? [6]
c) What is interface in java? How to declare an interface, write syntax? Can we achieve multiple inheritance by using interface? [6]

P.T.O.

- Q5)** a) What is RuntimeException? Demonstrate with the example, how to handle the Number Format exception? [6]
- b) Explain try, catch and finally with suitable example. [6]
- c) What is Collection? Explain set interface. [6]

OR

- Q6)** a) What is checked exception? Demonstrate Arithmetic Exception using throw. [6]
- b) What is Generics? Write a code of ArrayList using generics. [6]
- c) Differentiate between ArrayList and LinkedList collection classes. [6]

- Q7)** a) Explain Concept of Stream with its types. [6]
- b) Explain singleton pattern with example. [6]
- c) Write the Java program to read a file using FileInputStream class. [5]

OR

- Q8)** a) Explain Iterator Pattern with its merits. [6]
- b) Write a Program to write a file using FileOutputStream class. [6]
- c) What are design patterns? Explain 'Adaptor' design pattern in detail. [5]

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